



IT3060 – Human Computer Interaction

BSc (Hons) in Information Technology – Year 3

Lab Exercise 01

User Research & Requirements Gathering for the FitFlow App Redesign

Student Name: Bimsara P

Student ID: IT23629394

Group: Y3.S2.WE.IT.0202

Date of Submission: 2026/08/02

Activity 1: Stakeholder Identification

FitFlow is a fitness mobile application undergoing a redesign after a period of declining engagement. The redesign introduces AI-generated workout plans, nutrition/meal tracking, and social/community features such as challenges. Understanding who is affected by the product — directly and indirectly — is the first step of user research, since it determines who should be interviewed, surveyed, or observed, and whose priorities must be balanced during design decisions.

1.1 Stakeholder Map

Stakeholders were grouped into three categories based on their proximity to actual use of the FitFlow app: Primary (direct hands-on users), Secondary (people who influence, support, or are indirectly served through a primary user), and Tertiary (organizational or indirectly affected parties).

Primary Stakeholders (Direct Users)

Stakeholder	Description	Why Important to Research
Casual / everyday users	People who log occasional workouts and meals for general fitness or weight management.	Represent the largest user segment; their drop-off is likely the cause of declining engagement, so their pain points must shape the redesign.
Fitness enthusiasts / power users	Frequent users who train regularly and want advanced tracking analytics and personalization.	Heavy users generate most in-app data and are the earliest adopters of new features like AI coaching; their feedback validates advanced functionality.
Premium subscribers	Users who pay for advanced AI coaching, detailed nutrition plans or ad-free experience.	Revenue depends on this group; understanding what justifies payment is critical to increasing premium conversion.
New users / first-time downloaders	People onboarding to FitFlow for the first time.	Onboarding experience determines retention; their confusion points reveal usability issues before habits form.
Users with accessibility needs	Users with visual, motor, or cognitive impairments who rely on accessible design.	Ensures the app is inclusive and meets accessibility audits mentioned in the case study (final accessibility audits before launch).

Secondary Stakeholders (Influence / Support Primary Users)

Stakeholder	Description	Why Important to Research
Personal trainers / fitness coaches	Professionals whose clients use FitFlow, or who may use it to assign/monitor plans.	Provide expert insight into whether AI-generated workouts are realistic, safe, and trustworthy compared to human coaching.
Nutritionists / dietitians	Professionals who may recommend or review the meal-logging feature.	Validate whether nutrition-tracking data and suggestions are accurate and clinically sound.
Workout partners / friends / family	People who join community challenges with the primary user or provide social encouragement.	Social features (challenges, sharing) directly depend on how these secondary users interact with and influence primary users.
Customer support agents	Staff who handle user complaints, bug reports, and onboarding queries.	Hold aggregated insight into recurring pain points across thousands of users, useful for triangulating research findings.
Marketing team	Internal team responsible for campaigns and positioning (e.g., the soft-launch A/B testing mentioned in the case study).	Their messaging must align with what users actually value, so their input shapes which features to highlight in research and launch.

Tertiary Stakeholders (Indirect / Organizational)

Stakeholder	Description	Why Important to Research
Investors / company leadership	Stakeholders funding the redesign and expecting business outcomes (retention, ratings, revenue).	Research findings must be translated into metrics (SUS, retention, conversion) that justify continued investment.
App store platforms (Apple App Store / Google Play)	Platforms that review, rank and approve the app for distribution.	Compliance with platform guidelines (privacy, accessibility) affects whether design decisions can even ship.
Data protection regulators	Bodies governing user data privacy (relevant given on-device AI and cloud processing).	The privacy-focused architecture mentioned in the case study must be validated against regulatory expectations uncovered in research.
Competing fitness apps	Rival products competing for the same user base.	Understanding why users compare FitFlow unfavourably to competitors helps prioritise which pain points matter the most.
Healthcare / insurance partners	Organisations that might integrate with or reference fitness data (e.g., wellness incentive programmes).	Indirectly affected by data accuracy and sharing decisions made based on user research.

Overall, primary stakeholders define what the app must do well day-to-day, secondary stakeholders reveal how the app fits into a broader ecosystem of support and social influence, and tertiary stakeholders set the boundaries (business, legal, platform) within which the redesign must operate. Mapping all three ensures the research plan does not focus solely on end users while ignoring the constraints and influences that ultimately shape adoption and success.

Activity 2: Selection of Research Methods

2.1 Chosen Methods

- Semi-structured interviews — with regular users, power users and fitness trainers to explore motivations, trust in AI coaching, and detailed pain points.
- Online survey (questionnaire) — distributed to a broad sample of existing users to quantify satisfaction, feature demand, and usage frequency.
- Contextual inquiry / naturalistic observation — watching users perform real tasks (starting an AI plan, logging a meal, joining a challenge) in their own environment (e.g., at home or gym).

2.2 Justification

Three user research methods were selected because they complement one another by providing insights into the depth, breadth, and accuracy of user experiences and behaviours. Interviews provide rich qualitative data that can help identify the underlying reasons for users' frustrations with the existing application, as well as their expectations regarding the trustworthiness and usefulness of AI-based coaching. Such detailed insights may not be adequately captured through quantitative methods alone. Surveys, in contrast, enable data to be collected from a larger and more diverse sample of users using standardized questions. This produces quantifiable findings, such as the percentage of users interested in social challenges, which can be used to identify common user requirements and support feature prioritization. Contextual inquiry provides an additional perspective by examining users' actual interactions and behaviours within their normal context of use. This is particularly valuable because users may unintentionally overlook or underreport minor usability issues, such as abandoning meal logging midway due to the effort involved. By combining these three methods, the research can achieve triangulation, allowing findings to be compared across different sources of evidence. For example, if interviews, surveys, and contextual inquiry all indicate that meal logging is tedious or time-consuming, the development team can have greater confidence that this represents a genuine and high-priority usability issue rather than an isolated finding associated with a single research method.

2.3 Combining Qualitative and Quantitative Approaches

The survey will combine closed quantitative items (Likert scales, multiple-choice, usage frequency) with a small number of open-ended qualitative questions, allowing statistical analysis of satisfaction/priority scores alongside illustrative quotes. Interview and observation data will be qualitative, analyzed thematically (grouping recurring pain points such as "AI plan feels inflexible" or "hard to find nutrition log") and then cross-checked against the survey's quantitative results — for example, if 70% of survey respondents rate nutrition tracking as "difficult," the interview transcripts and observation notes can be reviewed for the specific usability breakdowns that explain that score. This mixed-methods approach ensures conclusions are both statistically representative and rich enough to inform concrete design decisions.

Activity 3: Research Instruments

3.1 Semi-Structured Interview Guide

Target participants: regular users, power users, and fitness trainers. Duration: 25–35 minutes. Setting: remote video call or in person, audio-recorded with consent.

Introduction (not scored, for rapport-building)

- Thank the participant, explain the purpose of the study, confirm consent and recording permission, remind them there are no right/wrong answers.

Main Questions (8–12)

1. Can you briefly describe how and why you first started using FitFlow?
2. Walk me through a typical week of how you use the app — which features do you open most often?
3. What is the most frustrating part of using FitFlow today?
4. Have you tried the AI-generated workout plans? What did you think of the recommendations?
5. How much do you trust an AI to adjust your workout plan without a human trainer reviewing it? Why?
6. Tell me about the last time you logged a meal. How easy or difficult was that process?
7. What information about nutrition would be most useful for the app to show you?
8. Have you used, or would you use, social/community features such as challenges with friends? Why or why not?
9. Can you describe a time FitFlow helped you feel motivated, and a time it did not?
10. If you could add one new feature to FitFlow, what would it be and why?
11. Is there anything about your privacy or data that concerns you when using AI-based fitness apps?
12. Is there anything else about your experience with FitFlow you'd like to share that we haven't discussed?

Closing

- Thank the participant, explain next steps, and offer to share findings/summary if requested.

3.2 Survey Questionnaire

Target: broad sample of existing FitFlow users. Delivery: online form (e.g., Google Forms), estimated completion time 5–7 minutes. Includes a mix of multiple-choice, Likert-scale, and open-ended questions (15 total).

#	Question	Type	Response Options
1	How long have you been using FitFlow?	Multiple Choice	Less than 1 month / 1–6 months / 6–12 months / Over 1 year
2	How often do you open the FitFlow app?	Multiple Choice	Daily / A few times a week / Weekly / Rarely
3	Overall, I am satisfied with the FitFlow app.	Likert Scale (1–5)	1 = Strongly Disagree, 5 = Strongly Agree

#	Question	Type	Response Options
4	Logging a meal in FitFlow is quick and easy.	Likert Scale (1–5)	1 = Strongly Disagree, 5 = Strongly Agree
5	I trust the AI-generated workout plans to suit my fitness level.	Likert Scale (1–5)	1 = Strongly Disagree, 5 = Strongly Agree
6	Have you used the AI workout planning feature?	Multiple Choice	Yes, regularly / Yes, tried once / No, but interested / No, not interested
7	Have you participated in a community challenge on FitFlow?	Multiple Choice	Yes / No, but would like to / No, not interested
8	Social/community features (challenges, sharing progress) make me more motivated.	Likert Scale (1–5)	1 = Strongly Disagree, 5 = Strongly Agree
9	Which feature is most important to you?	Ranking / Multiple Choice	AI workout plans / Nutrition tracking / Social challenges / Progress analytics
10	I find it easy to navigate between workouts, nutrition, and social sections of the app.	Likert Scale (1–5)	1 = Strongly Disagree, 5 = Strongly Agree
11	Are you currently a premium (paid) subscriber?	Multiple Choice	Yes / No / Considering it
12	What is the single biggest problem you experience while using FitFlow?	Open-ended	Free text
13	What new feature would most improve your experience?	Open-ended	Free text
14	I feel my personal data is handled safely and privately by FitFlow.	Likert Scale (1–5)	1 = Strongly Disagree, 5 = Strongly Agree
15	Any additional comments or suggestions?	Open-ended	Free text

3.3 Observation / Contextual Inquiry Checklist

Used during contextual inquiry sessions where the researcher observes the participant using FitFlow in their normal environment while completing three tasks: (1) set up an AI workout plan, (2) log a meal, (3) join a community challenge. The researcher records notes against each checklist item and times each task.

Onboarding

- ☐ Does the user complete sign-up without help?
- ☐ Does the user understand what permissions are being requested (data, notifications)?

AI Workout Setup

- ☐ Can the user set fitness goals without confusion?
- ☐ Does the user understand how the AI plan was generated?
- ☐ Does the user attempt to modify or override an AI suggestion?

Meal / Nutrition Logging

- ☐ How many steps does it take the user to log a meal?
- ☐ Does the user use search, barcode scan, or manual entry — and which do they prefer?
- ☐ Does the user abandon the task or show signs of frustration?

Social / Community Features

- ☐ Can the user find and join a challenge without guidance?
- ☐ Does the user understand what happens after joining (visibility, leaderboard, privacy)?
- ☐ Does the user comment on privacy concerns while using social features?

General Navigation

- ☐ Does the user get lost between the workout, nutrition, and social tabs?
- ☐ Does the user use any accessibility settings (text size, voice-over)?
- ☐ What verbal reactions (frustration, delight, confusion) does the user express?

For each task the observer also records: task completion (yes/partial/no), time on task, number of errors/hesitations, and any verbal think-aloud comments.

Activity 4: Pilot Testing of Research Instruments

The interview guide, survey questionnaire, and observation checklist were piloted with three classmates acting as proxy users before being used with actual participants, in order to check clarity, length, and difficulty.

4.1 Pilot Feedback Summary

Pilot Tester	Instrument(s) Tested	Feedback (clarity / length / difficulty)	Changes Made After Pilot
Pilot Tester 1	Interview guide, Survey	Q5 (AI trust) felt repetitive with Q4; survey took 9 minutes, longer than expected.	Merged Q4 and Q5 into one combined question; shortened two open-ended survey questions to reduce fatigue.
Pilot Tester 2	Survey, Observation checklist	Some Likert questions did not specify what "agree" meant in context (e.g., 'easy to navigate' was vague about which screen).	Added examples in brackets to clarify ambiguous Likert items (e.g., specifying 'workout, nutrition, and social tabs').
Pilot Tester 3	Interview guide, Observation checklist	Interview question about privacy felt abrupt near the end; observation checklist lacked a place to note time-on-task.	Reordered privacy question earlier in the flow with softer phrasing; added explicit time-on-task and error-count fields to the checklist.

4.2 Summary of Pilot Testing Process and Outcomes

Each of the three pilot testers completed the survey independently, then took part in a short mock interview and a brief walkthrough of the observation checklist while the researcher practiced notetaking. On average, the original survey took 8–9 minutes to complete, slightly above the 5–7-minute target, and the original interview guide contained two overlapping questions on AI trust. Feedback indicated the instruments were generally clear, but a small number of questions were ambiguous or repetitive, and the observation checklist lacked space to capture timing data. Following the pilot, questions were merged or clarified, ambiguous scale items were given concrete examples, one sensitive question was sequenced, and explicit timing/error fields were added to the observation checklist. After revision, a fourth informal read-through with one of the same testers confirmed the survey could be completed in under 7 minutes and that no confusing items remained, giving confidence that the instruments were ready for use with real research participants.

Activity 5: User Research with Participants

Research was conducted with five participants spanning three stakeholder groups (regular/casual users, a power user, a premium subscriber, a fitness trainer, and a first-time user) to ensure the findings reflect more than one perspective, as required.

ID	Stakeholder Group	Profile Summary	Method(s) Used
P1	Regular / casual user	3 months using FitFlow, exercises 2–3x/week	Interview + Survey
P2	Power user / fitness enthusiast	Uses FitFlow daily, trains 5x/week	Interview + Observation
P3	Premium subscriber	Pays for AI coaching plan, 1 year on FitFlow	Survey + Observation

ID	Stakeholder Group	Profile Summary	Method(s) Used
P4	Fitness trainer (secondary stakeholder)	Certified personal trainer, some clients use FitFlow	Interview
P5	New user (first-time)	Downloaded FitFlow this week for a trial	Observation + Survey

5.1 Participant Overview

5.2 Key Findings and Raw Notes Summary

ID	Group	Key Raw Notes / Observations	Implication for Design
P1	Regular user	Meal logging via manual search felt slow (~6 taps to log a snack); liked the AI plan but wanted to swap individual exercises easily.	Simplify meal search/autocomplete; add per-exercise swap option in AI plans.
P2	Power user	Wanted more detailed analytics and the ability to fully override AI-suggested intensity; found social challenge leaderboard motivating.	Add manual override controls for AI plans; keep and expand leaderboard/community features.
P3	Premium subscriber	Felt the AI coaching justified the subscription but was unsure why certain nutrition targets were recommended (no explanation shown).	Add short rationale/explanation text next to AI nutrition recommendations to build trust.
P4	Fitness trainer	Cautious about clients following AI plans without professional oversight, especially around injury history; suggested a way to flag/import client medical constraints.	Consider optional trainer-review or constraint-input step before AI plan finalization.
P5	New user	Got lost during onboarding trying to find where to join a challenge; joined a challenge accidentally without understanding visibility settings.	Clarify onboarding flow and add an explicit privacy/visibility confirmation step when joining challenges.

5.3 Overall Themes

- Meal logging is perceived as slow or tedious across most participants, regardless of experience level.
- Trust in AI recommendations increases when users are given an explanation and the ability to override or adjust suggestions.
- Trainers see value in AI coaching but want a safeguard for injury/medical constraints.
- Social/community features are motivating once discovered, but new users can join challenges without realizing privacy implications.
- These themes are consistent with the case study's later usability testing findings, which also identified the need for 'more user control over AI suggestions and simplifying social navigation.'

Activity 6: Consent Form and Data Management

6.1 Participant Consent Form

FitFlow User Research — Participant Consent Form

Study Purpose

This research is being conducted as part of IT3060 – Human Computer Interaction (Lab Exercise 01) to understand how people currently use the FitFlow fitness app and to gather requirements for a redesign involving AI-based workout coaching, nutrition tracking, and social/community features. Your input will help identify pain points and inform design recommendations.

Voluntary Participation

Your participation in this study is entirely voluntary. You may decline to answer any question and may withdraw from the session at any time without giving a reason and without any negative consequence.

What Participation Involves

You will be asked to take part in an interview, complete a short survey, and/or be observed while performing simple tasks in the FitFlow app (such as setting up an AI workout plan, logging a meal, or joining a challenge). The session will take approximately 20–35 minutes.

Recording Permission

With your permission, the session may be audio recorded (and, for observation sessions, the screen may be recorded) solely to help the researcher accurately capture your feedback. Recordings will not be shared outside the research team.

- ☐ I agree to audio/screen recording of this session.
- ☐ I do NOT agree to recording; I understand the researcher will instead take written notes.

Data Usage and Confidentiality

Your responses will be anonymized (no names or other directly identifying details will appear in the report) and used only for the purposes of this academic lab exercise. Data will not be shared with third parties and will not be used for any commercial purpose.

Right to Withdraw

You may withdraw your consent and request that your data be excluded from the study at any point up until the findings are finalized, by contacting the researcher using the details below.

Confidentiality

All raw notes, recordings, and survey responses will be stored securely and only accessible to the researcher(s) conducting this study, as described in Section 6.2 below.

Consent Declaration

- ☐ I have read and understood the information above.
- ☐ I voluntarily agree to take part in this research.
- ☐ I understand I can withdraw at any time.

Participant Name (or ID): _____ Signature: _____ Date: _____

Researcher Name: _____ Signature: _____ Date: _____

6.2 Raw Data Storage, Anonymization, and Security

Storage

- Interview audio recordings and notes are stored in a password-protected folder on the researcher's personal cloud drive, accessible only to the research team for this lab exercise.
- Survey responses (collected via an online form) are stored in the form provider's response sheet, with access restricted to the researcher's account.
- Observation checklists and timing data are recorded on paper/digital checklist templates and scanned or transcribed into a shared research log soon after each session.

Anonymization

- Each participant is assigned a code (P1–P5) at the time of the session; names are never recorded alongside responses in the analysis file.
- Any identifying details mentioned during interviews (e.g., workplace, exact age) are removed or generalized in the written summary (e.g., "works in an office job" instead of a specific employer).
- A single master list linking participant codes to real identities is kept separately from the research data and will be deleted once grading/verification is complete.

Security Measures

- All digital files are stored on password-protected devices/accounts; recordings and the master identity list are not uploaded to public or shared links.
- Only the research team involved in this lab exercise has access to raw data; data is retained only as long as needed for the assignment and then deleted.
- Printed consent forms and checklists are kept in a physically secure location (e.g., a locked folder) until the exercise is graded.